School of Computer Science and Artificial Intelligence

Lab Assignment # 04

Course Title : **AI Assistant Coding**
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Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques

Task 1: Zero-Shot Prompting – Leap Year Check

❖ ***Scenario: Zero-shot prompting involves giving instructions without providing examples.***

Task Description: ***Use zero-shot prompting to instruct an AI tool to generate a Python function that:***

- ***Accepts a year as input***
- ***Checks whether the given year is a leap year***
- ***Returns an appropriate result***

Note: No input-output examples should be provided in the prompt.

Expected Output

- ***AI-generated leap year checking function***
- ***Correct logical conditions***
- ***Sample input and output***
- ***Screenshot of AI-generated response (if required)***

• Prompt used:

Write a Python program that accepts a year as input, checks whether the given year is a leap year, and returns an appropriate result.

• Input:

2020
2025
2014

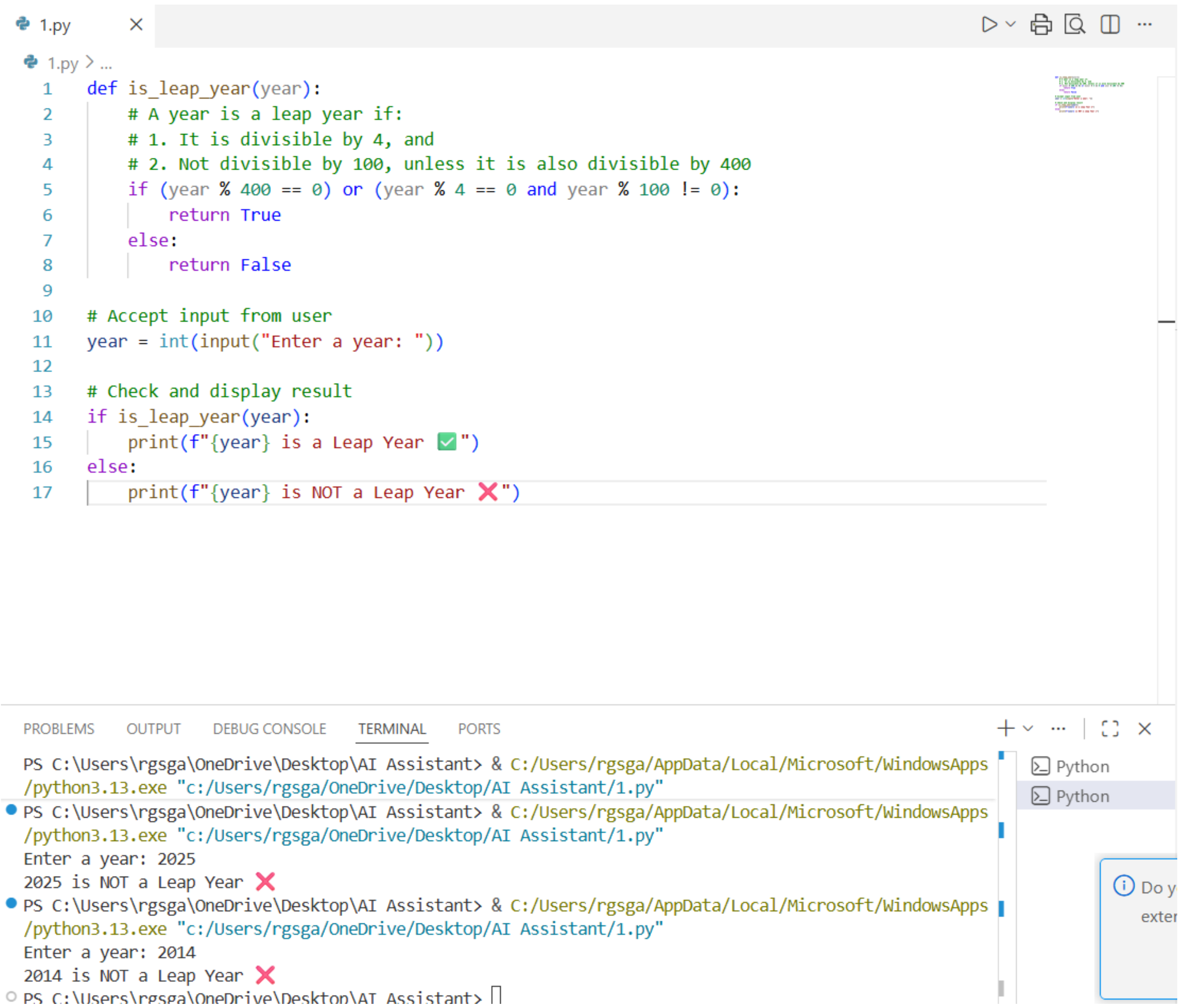
• Output:

Enter a year: 2020
2020 is a Leap Year

Enter a year: 2025
2025 is NOT a Leap Year

Enter a year: 2014
2014 is NOT a Leap Year

• Screenshot of Generated Code:



• Short Explanation of Code:

The code checks if a year is a leap year by testing divisibility: divisible by 400, or divisible by 4 but not by 100, then prints the result accordingly.

Task 2: One-Shot Prompting – Centimeters to Inches Conversion

❖ ***Scenario: One-shot prompting guides AI using a single example.***

Task Description: ***Use one-shot prompting by providing one input-output example to generate a Python function that:***

- ***Converts centimeters to inches***
- ***Uses the correct mathematical formula***

Example provided in prompt: ***Input: 10 cm → Output: 3.94 inches***

Expected Output

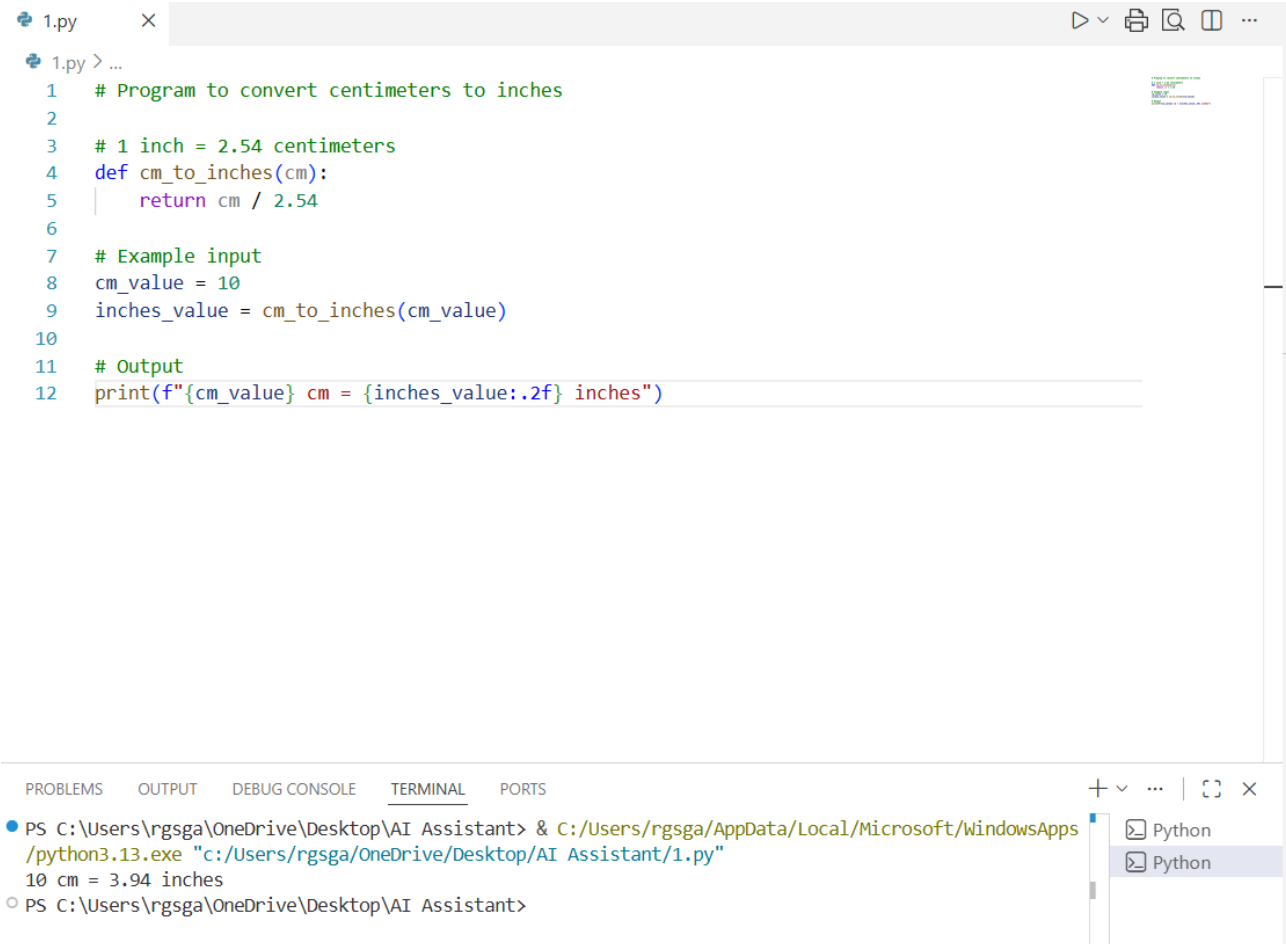
- ***Python function with correct conversion logic***
- ***Accurate calculation***
- ***Sample test cases and outputs***

• Prompt used:
Write a Python program that converts centimeters to inches using the formula inches = centimeters / 2.54. For example, if the input is 10 cm, the output should be 3.94 inches.

• Input:
10 cm

• Output:
3.94 inches

• Screenshot of Generated Code:



• Short Explanation of Code:
It converts centimeters to inches using the formula inches = centimeters / 2.54. It defines a function cm_to_inches(cm) that performs the conversion, takes an input value (like 10 cm), and prints the result formatted to two decimal places. For example, 10 cm becomes 3.94 inches.

Task 3: Few-Shot Prompting – Name Formatting

❖ ***Scenario: Few-shot prompting improves accuracy by providing multiple examples.***

Task Description: ***Use few-shot prompting with 2–3 examples to generate a Python function that:***

- ***Accepts a full name as input***
- ***Formats it as "Last, First"***

Example formats:

- ***"John Smith" → "Smith, John"***
- ***"Anita Rao" → "Rao, Anita"***

Expected Output

- ***Well-structured Python function***
- ***Output strictly following example patterns***
- ***Correct handling of names***
- ***Sample inputs and outputs***

• Prompt used:

Write a Python function that accepts a full name and returns it in the format "Last, First" for the given inputs John Smith – Smith John and Anith Rao – Rao Anitha.

• Screenshot of Generated Code:

1.py

1.py > ...

```python
def format_name(full_name: str) -> str:
    """
    Accepts a full name and returns it in the format 'Last, First'.
    Assumes input is in 'First Last' format.
    """
    parts = full_name.strip().split()
    if len(parts) >= 2:
        first, last = parts[0], parts[-1]
        return f"{last}, {first}"
    else:
        return full_name  # return unchanged if only one part

# --- Sample Inputs and Outputs ---
print(format_name("John Smith"))     # Output: Smith, John
print(format_name("Anita Rao"))      # Output: Rao, Anita
print(format_name("Michael Jordan")) # Output: Jordan, Michael
print(format_name("Madonna"))        # Output: Madonna
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assistant/1.py"
Smith, John
Rao, Anita
Jordan, Michael
Madonna
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant>
```

Python

Python

• Short Explanation of Code:

The code splits the full name into parts, takes the last word as the **last name**, and everything before it as the **first name(s)**, then returns them in the format "Last, First". If only one word is given, it just returns that unchanged.

Task 4: Comparative Analysis – Zero-Shot vs Few-Shot

❖ ***Scenario**: Different prompt strategies may produce different code quality..*

Task Description:

- ***Use zero-shot prompting to generate a function that counts vowels in a string***
- ***Use few-shot prompting for the same problem***
- ***Compare both outputs based on: o Accuracy o Readability o Logical clarity***

Expected Output

- ***Two vowel-counting functions***
- ***Comparison table or short reflection paragraph***
- ***Conclusion on prompt effectiveness***

• Prompt used:

◇ Zero-Shot Prompt

Prompt: *"Write a Python function to count vowels in a string."*

◇ Few-Shot Prompt

Prompt: *"Here are examples of functions that process strings:*
Example 1: Count consonants in a string.
Example 2: Count digits in a string. Now, write a function to count vowels in a string."

• Screenshot of Generated Code:

1.py

```python
def count_vowels(text):
    vowels = "aeiouAEIOU"
    count = 0
    for char in text:
        if char in vowels:
            count += 1
    return count

# Test cases
print(count_vowels("Hello World"))
print(count_vowels("Data Science"))
print(count_vowels("AEIOU"))

#Zero Short
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assistant/1.py"
3
5
```

```python
def count_vowels(text):
    vowels = set("aeiouAEIOU")
    return sum(1 for char in text if char in vowels)

# Test cases
print(count_vowels("Hello World"))
print(count_vowels("Data Science"))
print(count_vowels("AEIOU"))

# Few Short
```

```
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/rgsga/OneDrive/Desktop/AI Assistant/1.py
3
5
5
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant>
```

Comparison Table

Aspect	Zero-Shot Output	Few-Shot Output
Accuracy	Correctly counts vowels	Correctly counts vowels
Readability	Clear but slightly verbose (loop + counter)	More concise, uses `sum()` with generator
Logical Clarity	Straightforward, beginner-friendly	Compact, Pythonic, shows awareness of idiomatic style

• Short Explanation of Code:

Zero-Shot Code: The zero-shot function defines a string of vowels and uses a loop to check each character in the input. If the character is a vowel, a counter increases. Finally, the function returns the total count. This makes the logic very clear and easy to follow step by step.

Few-Shot Code: The few-shot function creates a set of vowels and uses a generator expression inside sum(). Each vowel found contributes one to the total, which is returned directly. This approach is shorter, cleaner, and more Pythonic.

• Conclusion:

Few-shot prompting tends to produce **cleaner and more idiomatic code**, while zero-shot prompting gives a **basic but reliable solution**. For teaching or clarity, zero-shot is better; for efficiency and elegance, few-shot wins.

Task 5: Few-Shot Prompting – File Handling

❖ ***Scenario**: File processing requires clear logical understanding.*

Task Description: *Use few-shot prompting to generate a Python function that:*

- ***Reads a .txt file***
- ***Counts the number of lines in the file***
- ***Returns the line count***

Expected Output

- ***Working Python file-processing function***
- ***Correct line count***
- ***Sample .txt input and output***
- ***AI-assisted logic explanation***

• Prompt used:

Write a python function that reads a .txt file, counts the number of lines, and returns the line count.

Example 1: Input file with 3 lines → Output: 3

• Screenshot of Generated Code:

• Input:

Hello world
This is a test file
It has three lines

• Output: **Number of lines in sample.txt: 3**

•AI-Assisted Logic Explanation:

- **Step 1:** Open the file safely using with open(...) so it auto-closes.
- **Step 2:** Read all lines into a list with readlines().
- **Step 3:** Use len(lines) to count how many lines exist.
- **Step 4:** Return that count.
- **Step 5:** Handle errors like missing files with try-except.